



FACULTY OF COGNITIVE SCIENCE AND HUMAN DEVELOPMENT

KMK 3013 KNOWLEDGE-BASED SYSTEM

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TITLE: DESIGNING AN AGENTIC AI FOR ADHD TASK MANAGEMENT

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1.0 Introduction

The interviewee involved in this project is Darren Sebastian, a registered counselor under Lembaga Kaunselor Malaysia with experience in supporting individuals with ADHD-related behavioral and task management challenges. The interview was conducted to gather expert knowledge and practical insights regarding ADHD-related behaviors, challenges, and strategies that can help individuals manage tasks more effectively.

Individuals with ADHD often face difficulties in maintaining focus, organizing tasks, and completing long or complex activities. These challenges can affect productivity and may lead to stress, procrastination, or difficulty managing daily responsibilities. Therefore, understanding effective strategies from expert experience is important in developing a supportive system that can assist users in task management.

The information gathered from the interview was analyzed and converted into structured knowledge for the AI assistant system. The extracted knowledge was later represented as declarative knowledge, procedural knowledge, and system constraints to ensure that the system follows expert-defined guidance. In addition, the GDVRR (Generate, Deconstruct, Verify, Rewrite, Reflect) protocol was applied to evaluate the AI system and improve its reasoning process.

This project aims to demonstrate how expert knowledge can be integrated into an AI-based system to provide more structured, supportive, and effective assistance for individuals with ADHD.

2.0 Human Expert Knowledge Elicitation

2.1 Interview Background

A semi-interview was conducted with Darren Sebastian, a registered counselor under Lembaga Kaunselor Malaysia, to gather information related to ADHD task management strategies. The interview session lasted approximately 5-10 minutes and focused on understanding the challenges faced by individuals with ADHD, as well as effective methods that can help improve focus and reduce overwhelm during task completion.

The interview was conducted in an informal and conversational manner to encourage natural responses. The information obtained from the interview was later analyzed and converted into structured rules and system design components for the AI assistant.

2.2 Interview transcript

Q1: How do people with ADHD usually handle long tasks?

Interviewee:

People with ADHD usually struggle to complete long tasks in one sitting because they have difficulty maintaining focus on the task. As counselors, we usually divide the task into smaller parts so it becomes easier and more manageable for them compared to doing one long task continuously.

Q2: What happens if a task feels too long or complicated?

Interviewee:

Like most people, they may start procrastinating or avoid the task because it feels stressful and difficult to manage. Long and complicated tasks can increase pressure, especially for individuals with certain challenges such as ADHD.

Q3: What makes tasks overwhelming for individuals with ADHD?

Interviewee:

Too many instructions at once can create confusion and emotional stress. Usually, we give short explanations and simple instructions step-by-step. For example, do task A first, then task B. If the instructions are too long, they may struggle to follow them properly.

Q4: If someone with ADHD feels overwhelmed, what should they do first?

Interviewee:

First, we try to help them calm down and relax. After that, we guide them to focus on one small and simple step first. Instead of rushing them, we slowly guide them through the task to reduce pressure and make the task feel easier and more enjoyable.

Q5: What kind of instructions work best?

Interviewee:

Short, clear, and step-by-step instructions work best. The instructions need to be simple and structured so they can understand and follow them more effectively.

Q6: What strategies help improve focus?

Interviewee:

One useful strategy is using a timer. For example, studying for 25 minutes followed by a short break can help maintain attention and reduce distractions.

Q7: How can someone manage distractions while doing work?

Interviewee:

Reducing distractions, setting reminders, and focusing on one task at a time can improve concentration. When they focus on a task for a short period of time, they are usually able to concentrate better.

Q8: How do people with ADHD usually organize tasks?

Interviewee:

We usually help them create checklists or to-do lists so tasks can be organized more clearly and progress becomes easier to track.

Q9: How should an AI assistant communicate with users with ADHD?

Interviewee:

The AI assistant should be supportive, encouraging, and avoid giving too many instructions at once.

Q10: Is there anything the assistant should avoid doing?

Interviewee:

The assistant should avoid overwhelming the user and should not provide medical advice or diagnosis.

2.3 Raw Extracted Knowledge

Interview Response	Extracted Rule / Knowledge
“People with ADHD usually struggle to complete long tasks in one sitting.”	If a task is long, it should be broken into smaller steps.
“Long or complicated tasks may increase procrastination and stress.”	Long tasks may increase avoidance behavior and stress.
“Too many instructions at once can create confusion.”	Instructions should be short, simple, and easy to follow.
“Starting with one small step helps reduce pressure.”	If the user feels overwhelmed, begin with one or two simple steps first.
“Using timers can help maintain attention and reduce distractions.”	If the user is distracted or struggling to focus, suggest a timer.
“Checklists or to-do lists help organize tasks better.”	Use checklist-based organization to support task management.
“The assistant should be supportive and avoid overwhelming the user.”	The AI assistant should use supportive language and avoid giving too many instructions at once.
“The system should not provide medical advice or diagnosis.”	The system must not provide medical advice or diagnosis.

Table 1: Raw Extracted Knowledge from Interview

3.0 Extracted Knowledge and System Design

3.1 Declarative knowledge (Knowledge Base)

Based on the interview responses from the registered counselor, several rules were extracted and converted into declarative knowledge for the AI system.

The extracted rules:

1. IF a task is long or contains many activities, THEN break the task into smaller and manageable steps.
2. IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first.

3. IF the instructions given are too long or complicated, THEN simplify them into short, clear, and easy-to-follow instructions.
4. IF too many instructions are presented at once, THEN reduce the number of instructions to prevent confusion and cognitive overload.
5. IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders.
6. IF the user requires emotional support or motivation, THEN provide supportive and encouraging responses.
7. IF the user has difficulty organizing tasks or tracking progress, THEN suggest using reminders, checklists, or to-do lists.
8. IF a task is long or highly complicated, THEN recognize that it may increase procrastination, stress, and task avoidance behavior.
9. IF the user requests medical diagnosis or treatment advice, THEN the system must not provide medical advice or diagnosis.

3.2 Procedural Knowledge (Tools)

Based on the strategies suggested during the interview, several procedural tools were identified for the AI assistant.

Tool Name	Function
Task_Chunking_Tool	Break large tasks into smaller and more manageable parts
Focus_Timer_Tool	Suggest structured focus sessions, such as 25-minute work periods followed by a short breaks
Checklist_Generator_Tool	Creates step-by-step task checklists
Reminder_Tool	Provides reminders for tasks and deadlines
Motivation_Tool	Generate supportive and encouraging messages

Table 2: Procedural Knowledge (AI Tools)

3.3 System Prompt

The AI assistant is guided by the following system constraints and prompt template:

You are an ADHD task management assistant. So your role is to help users manage tasks, reduce overwhelm, and improve focus. You must give short, clear, and step-by-step instructions. If a task is long, break it into smaller parts. If the user feels overwhelmed, start with only one or two simple steps. Use supportive and encouraging language. If the user is distracted, suggest timers or reminders. Do not provide medical advice or diagnosis.

3.4 System Architecture

The human interview knowledge was mapped into the AI system through three layers. Declarative knowledge was converted into IF–THEN rules, procedural knowledge was represented as functional tools, and behavioral constraints were implemented through the system prompt.

Gemini/OpenAI was used to simulate the Agentic AI assistant. The AI system was instructed to produce visible execution traces showing the rule checked, tool selected, reason for tool use, and final response. These traces were later analyzed using the GDVRR protocol.

Human Knowledge Component	AI System Component
Declarative Knowledge (rules from interview)	Inserted into AI prompt as IF–THEN rules
Procedural Knowledge (strategies/actions)	Implemented as functional AI tools
System Constraints (tone and boundaries)	Implemented through system prompt

Table 3: System Architecture Mapping

4.0 Applying the GDVRR Protocol

4.1 Generate

The AI assistant was tested using a complex ADHD-related scenario to evaluate whether the system followed the expert-derived rules, tools, and system constraints. The GDVRR protocol was applied to analyze the AI's reasoning process, identify weaknesses, and improve the system.

Prompt template Used

“You are an ADHD task management assistant. So your role is to help users manage tasks, reduce overwhelm, and improve focus. You must give short, clear, and step-by-step instructions. If a task is long, break it into smaller parts. If the user feels overwhelmed, start with only one or two simple steps. Use supportive and encouraging language. If the user is distracted, suggest timers or reminders. Do not provide medical advice or diagnosis.”

Scenario

“I have a 3-hour group assignment due tomorrow night. I feel very overwhelmed and I don't know where to start. I also keep getting distracted by my phone.”

Visible Execution Trace

Rule checked:

- Stage 1 (Recipient/Intent): Self-focused action.
- Stage 2 (Radioactive Vault): User expressed overwhelm but no sensitive categories were triggered.
- Stage 3 & 4 (Domain & Accuracy): Applied ADHD assistant constraints such as short instructions, task breakdown, and timer suggestions.

Strategy selected:

- Micro-tasking
- Environment Priming

Reason for strategy selection:

- When the brain feels "locked" due to overwhelm, the most effective strategy is to lower the barrier to entry by focusing on the physical environment and a single "non-scary" starting point.

AI Final Response

I hear you—that feeling of a looming deadline can make it hard to even open the laptop. We are going to handle this one tiny piece at a time. Take a deep breath.

Here are your first two steps:

Step 1: The "Phone Jail"

- **Action:** Put your phone in another room or inside a drawer.
- **Timer:** Set a timer for **20 minutes** on your computer or a kitchen timer. You only have to focus until it goes off.

Step 2: The 5-Minute Entry

- **Action:** Open the assignment document and read only the **first three requirements**.
- **Goal:** Do not try to write anything yet. Just read them so your brain knows what the target looks like.

You can do this. Once you've opened the doc and read those three lines, let me know and we'll pick the next smallest step.

4.2 Deconstruct

The AI assistant's execution trace was analyzed to understand how the system interpreted the user input and selected its reasoning process before generating the final response.

User Scenario

"I have a 3-hour group assignment due tomorrow night. I feel very overwhelmed and I don't know where to start. I also keep getting distracted by my phone."

Retrieved Information

From the user's prompt, the AI identified several important conditions:

- The task involved a long assignment requiring approximately three hours to complete.
- The task had an upcoming deadline due the next day.
- The user expressed feelings of overwhelm and stress.
- The user indicated difficulty starting the task.
- The user reported distractions caused by their phone.

AI Execution Trace Analysis

The AI generated the following reasoning stages:

- Rule Checked: Stage 1 (Recipient/Intent)

The AI classified the situation as a self-focused action and continued processing the request.

- Rule Checked: Stage 2 (Radioactive Vault)

The AI detected that the user expressed feelings of being overwhelmed. However, no sensitive or restricted categories such as medical diagnosis or crisis intervention were triggered.

- Rule Checked: Stage 3 & 4 (Domain & Accuracy)

The AI applied the predefined ADHD assistant constraints:

- provide short instructions
- break large tasks into smaller steps
- suggest timers to improve focus

Strategies Selected

The AI selected:

- Micro-tasking
- Environment Priming

AI Reasoning

“When the brain feels locked due to overwhelm, the most effective strategy is to lower the barrier to entry by focusing on the physical environment and a single non-scary starting point.”

The Micro-tasking strategy attempted to reduce the assignment into smaller actions to reduce cognitive overload. Meanwhile, Environment Priming attempted to modify the surroundings by removing distractions such as the phone.

Actions Generated by AI

Step 1: Phone Jail

Action: Place the phone in another room or inside a drawer.

Timer: Set a timer for 20 minutes to maintain concentration.

Step 2: 5-Minute Entry

Action: Open the assignment document and read only the first three requirements.

Goal: Reduce pressure and create an easier starting point.

Tool Usage Interpretation

Although the AI did not explicitly call predefined tools, the generated actions appeared to imitate several procedural tools:

- Task_Chunking_Tool → reducing a large assignment into smaller steps
- Focus_Timer_Tool → suggesting a timer
- Motivation_Tool → providing emotional encouragement

The AI attempted to balance emotional support, distraction reduction, and task simplification while following the ADHD assistant constraints.

4.3 Verify

The AI response and execution trace were evaluated against the expert-derived declarative knowledge, procedural tools, and system constraints to determine whether the assistant behaved according to the intended ADHD support framework.

The AI successfully followed several important expert-derived rules:

- The AI provided supportive and encouraging language.
- The AI reduced the assignment into smaller starting actions to reduce overwhelm.
- The AI suggested a timer to improve focus.
- The AI provided short and step-by-step instructions rather than overwhelming the user with many tasks simultaneously.

These behaviors aligned with the original ADHD assistant constraints. However, several weaknesses were identified during the verification process.

KBS Anomaly Analysis

Although the AI generally followed the intended support behavior, the execution trace introduced reasoning structures and terminology that were never defined in the expert-derived knowledge base, procedural tools, or system prompt. The following generated reasoning stages appeared:

- Stage 1 (Recipient/Intent)
- Stage 2 (Radioactive Vault)
- Stage 3 & 4 (Domain & Accuracy)

In addition, the AI introduced strategy labels:

- Micro-tasking
- Environment Priming

These reasoning structures and strategies were not extracted from the human expert interview and were never included within:

- declarative knowledge
- procedural tools
- system architecture
- system constraints

Therefore, the following anomaly was identified:

Primary KBS Anomaly Identified

Epistemic Boundary Violation (Knowledge Hallucination)

The AI generated reasoning structures, labels, and internal execution stages that were not present within the expert-derived knowledge base. Instead of strictly following the human-defined framework, the AI independently introduced additional reasoning concepts. Although the final response remained helpful, the reasoning process exceeded the intended expert boundaries.

Verification Conclusion

Human auditing was necessary because the hallucinated reasoning structures were not immediately obvious from the final answer alone. While the generated response sounded supportive and useful, the execution trace revealed that the AI relied partly on internally generated probabilistic reasoning patterns rather than strictly following deterministic expert-defined logic. This demonstrated the

importance of human verification in maintaining alignment between AI behavior and human expert knowledge.

4.4 Rewrite

After identifying the Epistemic Boundary Violation during the verification stage, manual intervention was performed to improve the AI system and reduce hallucinated reasoning structures. The prompt template, declarative rules, and procedural knowledge were refined to ensure that the AI assistant would only use concepts that were explicitly defined by the human expert knowledge base.

Old Declarative Rule

1. IF a task is long or contains many activities, THEN break the task into smaller and manageable steps.
2. IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first.
3. IF the instructions given are too long or complicated, THEN simplify them into short, clear, and easy-to-follow instructions.
4. IF too many instructions are presented at once, THEN reduce the number of instructions to prevent confusion and cognitive overload.
5. IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders.
6. IF the user requires emotional support or motivation, THEN provide supportive and encouraging responses.
7. IF the user has difficulty organizing tasks or tracking progress, THEN suggest using reminders, checklists, or to-do lists.
8. IF a task is long or highly complicated, THEN recognize that it may increase procrastination, stress, and task avoidance behavior.
9. IF the user requests medical diagnosis or treatment advice, THEN the system must not provide medical advice or diagnosis.

OLD Prompt Template

“You are an ADHD task management assistant. So your role is to help users manage tasks, reduce overwhelm, and improve focus. You must give short, clear, and step-by-step instructions. If a task is long, break it into smaller parts. If the user feels overwhelmed, start with only one or two simple steps. Use supportive and encouraging language. If the user is distracted, suggest timers or reminders. Do not provide medical advice or diagnosis.”

Old Procedural Knowledge

Tool Name	Function
Task_Chunking_Tool	Break large tasks into smaller and more manageable parts
Focus_Timer_Tool	Suggest structured focus sessions, such as 25-minute work periods followed by short breaks
Checklist_Generator_Tool	Creates step-by-step task checklists
Reminder_Tool	Provides reminders for tasks and deadlines
Motivation_Tool	Generates supportive and encouraging messages

Problem Identified

Although the AI produced a generally helpful response, the execution trace introduced reasoning structures and strategy labels that were not present within the expert-derived knowledge base.

Examples included:

- Stage 1 (Recipient/Intent)
- Stage 2 (Radioactive Vault)
- Stage 3 & 4 (Domain & Accuracy)
- Micro-tasking
- Environment Priming

These concepts were never defined within the declarative rules, procedural knowledge, system architecture, or system constraints. As a result, the AI exceeded the intended expert-defined framework and generated hallucinated reasoning structures.

The user scenario remained unchanged and continued to be used as the test input during the Generate phase. Only the declarative rules, prompt template, and procedural knowledge were revised to address the identified anomaly. This ensured that the effects of the modifications could be evaluated using the same testing scenario while maintaining consistency throughout the GDVRR process.

New Declarative Rule

10. IF the AI assistant generates reasoning structures, labels, or execution logic not explicitly defined in the expert-derived knowledge base, THEN reject and restrict them from being used.

New Adjusted Prompt Template

“You are an ADHD task management assistant. Your role is to help users manage tasks, reduce overwhelm, and improve focus. You must ONLY use reasoning structures, terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design. Do not invent new execution stages, labels, or strategy names.

Follow these expert-derived rules only:

1. IF a task is long or contains many activities, THEN break the task into smaller and manageable steps.
2. IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first.
3. IF the instructions given are too long or complicated, THEN simplify them into short, clear, and easy-to-follow instructions.
4. IF too many instructions are presented at once, THEN reduce the number of instructions to prevent confusion and cognitive overload.
5. IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders.
6. IF the user requires emotional support or motivation, THEN provide supportive and encouraging responses.
7. IF the user has difficulty organizing tasks or tracking progress, THEN suggest using reminders, checklists, or to-do lists.
8. IF a task is long or highly complicated, THEN recognize that it may increase procrastination, stress, and task avoidance behavior.
9. IF the user requests medical diagnosis or treatment advice, THEN do not provide medical advice or diagnosis.
10. IF the AI assistant generates reasoning structures, labels, or execution logic not explicitly defined in the expert-derived knowledge base, THEN reject and restrict them from being used.

Available procedural tools:

Task_Chunking_Tool: Break large tasks into smaller and manageable actions.

Focus_Timer_Tool: Suggest structured focus sessions.

Checklist_Generator_Tool: Create step-by-step task checklists.

Reminder_Tool: Provide reminders.

Motivation_Tool: Generate supportive and encouraging responses.

Before giving the final response, produce a visible execution trace using only this format:

Rule checked: state the activated rules

Tool selected: select tools only from the provided tool list

Reason for tool use: explain why the tools were selected

Then provide the final response.

Scenario: I have a 3-hour group assignment due tomorrow night. I feel very overwhelmed and I do not know where to start. I also keep getting distracted by my phone.

New Procedural Knowledge

Tool Name	Function
Task_Chunking_Tool	Break large tasks into smaller and manageable actions while preventing cognitive overload
Focus_Timer_Tool	Suggest structured focus sessions to improve concentration and reduce distractions
Checklist_Generator_Tool	Create step-by-step task checklists for easier task tracking
Reminder_Tool	Provide reminders for tasks and deadlines
Motivation_Tool	Generates supportive and encouraging responses to reduce emotional stress

Updated Visible Execution Trace

Rule checked:

- IF a task is long or contains many activities, THEN break the task into smaller and manageable steps.
- IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first.
- IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders.
- IF the user requires emotional support or motivation, THEN provide supportive and encouraging responses.
- IF a task is long or highly complicated, THEN recognize that it may increase procrastination, stress, and task avoidance behavior.

Tool selected:

- Task_Chunking_Tool
- Focus_Timer_Tool
- Motivation_Tool

Reason for tool use:

The user is facing a long, complex task due very soon, which is causing stress and potential task avoidance. Because the user feels overwhelmed, only a small number of starting steps should be provided to prevent cognitive overload. The user also reports distraction from their phone, making a structured focus strategy appropriate. Motivation_Tool was selected to provide encouragement and reduce anxiety.

New Improved AI Response

I completely understand why you feel overwhelmed right now. Facing a 3-hour assignment with a tight deadline is a lot of pressure, and it makes sense that distractions such as your phone become more appealing when stress increases.

You do not have to finish the entire assignment right now. We are only focusing on getting started.

Step 1: Put your phone in another room or out of sight and set a timer for 15 minutes.

Step 2: Open the assignment document and read only the instructions. Do not start writing yet. Simply focus on understanding what is required.

You can absolutely do this. Just 15 minutes. Let me know when your timer goes off, and we will figure out the next tiny step together!

Rewrite Outcome

The revised system successfully removed hallucinated execution stages and unsupported strategy labels that were previously generated by the AI. The updated execution trace only used expert-derived rules and predefined procedural tools. Furthermore, the revised response addressed both task overwhelm and phone-related distractions, demonstrating stronger alignment with the expert-defined ADHD support framework.

4.5 Reflect

The GDVRR analysis demonstrated that the AI assistant was generally capable of generating supportive and structured responses for ADHD-related task management. The system successfully reduced large tasks into smaller actions, used supportive language, and attempted to reduce distractions through timer-based strategies.

However, the verification process revealed that the original AI system could generate reasoning structures and terminology that were not present within the expert-derived knowledge base.

During the initial Generate phase, the AI independently introduced execution stages such as:

- Stage 1 (Recipient/Intent)
- Stage 2 (Radioactive Vault)
- Stage 3 & 4 (Domain & Accuracy)

The AI also generated strategy labels including:

- Micro-tasking
- Environment Priming

These concepts were not extracted from the human expert interview and were never defined in the declarative rules, procedural tools, or system architecture. This demonstrated that although the final

response appeared helpful, the AI relied partly on probabilistic reasoning patterns rather than strictly following deterministic expert-defined logic.

Large language models generate outputs based on learned probability patterns from training data. As a result, the AI attempted to introduce reasoning structures that seemed logically appropriate but were outside the intended expert framework. In contrast, the human expert relied on deterministic knowledge, where specific IF–THEN rules and predefined tools controlled system behavior.

Human verification was therefore essential because the hallucinated reasoning structures were not immediately obvious from the final response. Without auditing the execution trace, users might assume that the AI followed the expert system correctly simply because the generated output sounded convincing and supportive.

After manual intervention during the Rewrite stage, the system became more structured and aligned with the intended expert-defined framework. The refined prompt template successfully restricted the AI to predefined rules and procedural tools. The updated execution trace removed previously hallucinated stages and instead relied only on expert-derived knowledge. The revised system also selected predefined procedural tools such as `Task_Chunking_Tool`, `Focus_Timer_Tool`, and `Motivation_Tool`, allowing it to address both task overwhelm and phone-related distractions while remaining consistent with the expert-defined ADHD support framework.

This project also highlighted the risks of cognitive offloading, where users may become overly dependent on AI-generated responses without critically evaluating the reasoning process behind them. If users blindly trust AI-generated outputs without human auditing, hidden reasoning errors may remain undetected.

Overall, the GDVRR protocol demonstrated that AI systems should not function as unquestioned sources of authority. Human oversight remains necessary to ensure that Agentic AI systems remain aligned with human expert knowledge rather than relying entirely on probabilistic language generation.

GDVRR	Evaluation Metric	Student Analysis & Input
Generate	User Scenario and AI Output	Scenario used: “I have a 3-hour group assignment due tomorrow night. I feel very overwhelmed and I don’t know where to start. I also keep getting distracted by my phone.” The AI generated a response containing execution traces, tool selections, and step-by-step recommendations for task management.
Deconstruct	AI Execution Trace (Thought / Action)	The AI generated execution stages including Recipient/intent, Radioactive Vault, and Domain & Accuracy. It also selected the strategies Micro-tasking and Environment Priming before producing the final response. These reasoning stages and strategy labels were extracted from the execution trace and analyzed against the expert-derived knowledge base.
Verify	Human Expert Rule / System Constraints	The human expert stated that users who feel overwhelmed should be given only one or two simple starting steps, tasks should be broken into manageable parts, and the AI should only operate within the predefined expert-derived rules, tools, and constraints.
Verify	KBS Anomaly Identified	Epistemic Boundary Violation (Knowledge Hallucination). The AI generated reasoning stages and strategy labels that were not present in the declarative knowledge, procedural knowledge, system prompt, or system architecture derived from the expert interview.
Rewrite	Manual Correction / Prompt Engineering	The System Prompt was revised by adding the constraint: “You must ONLY use reasoning structures,

		terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design. Do not invent new execution stages, labels, or strategy names.” This modification was introduced to prevent unsupported reasoning structures from appearing in future outputs.
Reflect	Justification & Algorithmic Literacy	Human verification was necessary because the hallucinated reasoning structures appeared convincing despite not existing in the expert-derived knowledge base. This demonstrates how probabilistic language generation can produce plausible but unsupported reasoning. The GDVRR process helped identify these inconsistencies and ensured that the AI remained aligned with deterministic expert knowledge.

Table 4: GDVRR Audit Log

5.0 Individual Reflection

GDVRR Phase	Evaluation Metric	Student Analysis & Input
Deconstruct	AI's Execution Trace (Thought / Action)	Thought: "Stage 1 (Recipient/Intent) – Self-focused action." Action: The AI classified the user's request as a self-focused task and proceeded to the next reasoning stage before generating the response.
Verify	Human Expert Rule / System Constraint	IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first. The expert-derived knowledge base did not contain any reasoning stage called Recipient/Intent.
Verify	KBS Anomaly / Analysis Keyword	Epistemic Boundary Violation (Knowledge Hallucination)
Rewrite	Manual Correction / Prompt Engineering	I modified the System Prompt by adding the constraint: "You must ONLY use reasoning structures, terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design. Do not invent new execution stages, labels, or strategy names." Old Prompt: The original prompt guided the AI to reduce overwhelm but did not explicitly prevent the generation of new execution stages. New Prompt: The revised prompt now prohibits the creation of undefined execution stages.
Reflect	Justification & Algorithmic Literacy	This happened because AI usually generates responses based on probability patterns rather than strict expert rules. Even though Recipient/Intent sounded logical, it was not

		actually part of our original knowledge base. Human checking was important because if we only looked at the final answer, this issue could easily be missed. Without verification, users might trust AI outputs too much without checking whether the reasoning really follows expert knowledge.
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Table 5: Individual Reflection (Stella Francis 101115)

GDVRR Phase	Evaluation Metric	Student Analysis & Input
Deconstruct	AI's Execution Trace (Thought / Action)	Thought: “Stage 3 & 4 (Domain & Accuracy).” Action: The AI attempted to apply ADHD-related constraints before generating the final response.
Verify	Human Expert Rule / System Constraint	Declarative Knowledge, Procedural Knowledge, and System Constraints are the only components defined within the system architecture. No reasoning stage called Domain & Accuracy exists within the designed system architecture.
Verify	KBS Anomaly / Analysis Keyword	Epistemic Boundary Violation (Knowledge Hallucination)
Rewrite	Manual Correction / Prompt Engineering	I modified the System Prompt by adding the constraint: “You must ONLY use reasoning structures, terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design. Do not invent new execution stages, labels, or strategy names.” Old Prompt: The original prompt guided the AI to support ADHD task

		management but did not explicitly restrict additional reasoning frameworks. New Prompt: The revised prompt now prevents the AI from generating undefined reasoning stages.
Reflect	Justification & Algorithmic Literacy	The AI attempted to organize its reasoning process by creating a stage called Domain & Accuracy. However, this stage was not part of the architecture designed from the human expert interview. This demonstrates how AI may generate its own internal framework even when a deterministic structure has already been defined.

Table 6: Individual Reflection (Shirley Jane Julian 101023)

GDVRR Phase	Evaluation Metric	Student Analysis & Input
Deconstruct	AI's Execution Trace (Thought / Action)	Thought: "Strategy Selected: Micro-tasking." Action: The AI selected a strategy to reduce overwhelm by breaking the assignment into smaller tasks.
Verify	Human Expert Rule / System Constraint	IF a task is long or contains many activities, THEN break the task into smaller and manageable steps. Procedural Tool: Task_Chunking_Tool. The expert-defined procedural knowledge does not contain a strategy called Micro-tasking.
Verify	KBS Anomaly / Analysis Keyword	Epistemic Boundary Violation (Knowledge Hallucination)
Rewrite	Manual Correction / Prompt Engineering	I modified the System Prompt by adding the constraint: "You must ONLY use reasoning structures, terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design. Do not invent

		new execution stages, labels, or strategy names.” Old Prompt: The original prompt instructed the AI to break down tasks but did not explicitly prevent the generation of new strategy labels or reasoning structures. New Prompt: The revised prompt now requires the AI to use only reasoning structures and execution logic defined in the expert-derived knowledge base and prohibits the creation of new execution stages, labels, or strategy names.
Reflect	Justification & Algorithmic Literacy	The AI correctly recognized that the task should be broken down into smaller steps. However, instead of using terminology that existed in the expert-defined knowledge base, it generated a new strategy label called Micro-tasking. This shows how AI can create reasoning structures that sound reasonable but are not part of the deterministic expert framework. Human verification was important to ensure that the AI remained aligned with the expert-defined knowledge base.

Table 7: Individual Reflection (Vailery Keling Katan 101240)

GDVRR Phase	Evaluation Metric	Student Analysis & Input
Deconstruct	AI's Execution Trace (Thought / Action)	Thought: “Stage 2 (Radioactive Vault).” Action: The AI introduced an additional reasoning stage and checked whether the user’s request involved restricted or sensitive categories before continuing.

Verify	Human Expert Rule / System Constraint	IF the user requests medical diagnosis or treatment advice, THEN the system must not provide medical advice or diagnosis. The expert-derived knowledge base did not contain any reasoning stage called Radioactive Vault.
Verify	KBS Anomaly / Analysis Keyword	Epistemic Boundary Violation (Knowledge Hallucination)
Rewrite	Manual Correction / Prompt Engineering	I modified the System Prompt by adding the constraint: “You must ONLY use reasoning structures, terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design. Do not invent new execution stages, labels, or strategy names.” Old Prompt: The original prompt restricted medical advice but did not explicitly prevent the creation of additional reasoning stages. New Prompt: The revised prompt now prohibits undefined execution stages.
Reflect	Justification & Algorithmic Literacy	The AI created a reasoning stage that was never defined by the expert. Although the stage appeared meaningful, it was generated through probabilistic reasoning rather than deterministic expert logic. Human auditing was important because users would not normally notice this issue from the final answer alone.

Table 8: Individual Reflection (Nataly Kinli 100059)

GDVRR Phase	Evaluation Metric	Student Analysis and Input
Deconstruct	AI's Execution Trace (Thought / Action)	Thought: "Strategy Selected: Environment Priming." Action: The AI suggested reducing distractions by changing the user's surroundings before starting the task.
Verify	Human Expert Rule / System Constraint	IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders. Procedural Tools: Focus_Timer_Tool and Reminder_Tool. The expert-defined procedural knowledge does not contain a strategy called Environment Priming.
Verify	KBS Anomaly / Analysis Keyword	Epistemic Boundary Violation (Knowledge Hallucination)
Rewrite	Manual Correction / Prompt Engineering	I modified the System Prompt by adding the constraint: "Only use predefined procedural tools and do not invent new strategy labels." Old Prompt: The original prompt helped users manage distractions but did not explicitly prevent the creation of new strategy labels. New Prompt: The revised prompt now forces the AI to use predefined procedural tools from the expert knowledge base.
Reflect	Justification & Algorithmic Literacy	The AI identified that the user was distracted, but instead of using terminology that existed within the expert-defined knowledge base, it generated a new strategy called Environment Priming. Although the suggestion sounded useful, it was not part of the original expert framework. This demonstrates how AI can generate plausible reasoning structures through

		probabilistic language generation while operating outside deterministic expert knowledge.
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Table 9: Individual Reflection (Clare Brooklyn Anak Akei 101595)

6.0 Conclusion

This project demonstrated how expert knowledge can be integrated into an Agentic AI system to support ADHD-related task management. Through the interview with a registered counselor, valuable knowledge was collected and transformed into declarative rules, procedural tools, and system constraints that guided the behaviour of the AI assistant.

The GDVRR protocol was applied to evaluate the AI system and examine its reasoning process. Although the AI was able to provide supportive responses, break down tasks, and suggest useful focus strategies, the analysis revealed that it also generated reasoning structures that were not part of the expert-derived knowledge base. This issue was identified as an Epistemic Boundary violation and highlighted the importance of verifying AI-generated reasoning rather than only evaluating the final response.

After the Rewrite stage, improvements were made to the prompt template and knowledge base to ensure that the AI followed predefined rules and procedural tools more closely. The revised system produced more transparent and controlled reasoning that aligned better with the intended expert framework.

Overall, this project shows that AI can be a useful tool for supporting individuals with ADHD when it is guided by expert knowledge and monitored through human oversight. The findings also emphasize the risks of cognitive offloading, where users may become overly dependent on AI-generated outputs without evaluating the reasoning behind them. Therefore, human verification remains essential to ensure that AI systems remain reliable, trustworthy, and aligned with expert knowledge.

Appendix

Interviewee Voice Recording:

<https://drive.google.com/drive/folders/1hPLXuhygpPM9f7i1iGiMtEQNpI25fvX4?usp=sharing>

Original Agent Implementation Evidence:

The first screenshot shows the initial system prompt and scenario. The system prompt defines the role as an ADHD task management assistant, emphasizing short, clear, step-by-step instructions and supportive language. The scenario describes a user who is overwhelmed and distracted by their phone, needing help with a 3-hour group assignment due tomorrow night.

The second screenshot shows the execution trace starting with 'Rule Checked: Stage 1 (Recipient/Intent) - Self-focused action. Proceeded to Stage 2.' It continues with 'Rule Checked: Stage 2 (Radioactive Vault) - User expressed "overwhelmed," but no forbidden medical or sensitive categories were triggered. Proceeded to Stage 3.' and 'Rule Checked: Stage 3 & 4 (Domain & Accuracy) - Applied "ADHD Assistant" persona constraints: short steps, break down tasks, suggest timers.' The strategy selected is 'Micro-tasking and Environment Priming.' The reasoning states that when the brain feels 'locked' due to overwhelm, the most effective strategy is to lower the barrier to entry by focusing on physical environment and a single 'non-scary' starting point. The assistant then says: 'I hear you—that feeling of a looming deadline can make it hard to even open the laptop. We are going to handle this one tiny piece at a time. Take a deep breath. Here are your first two steps: Step 1: The "Phone Jail"'. The action is 'Put your phone in another room or inside a drawer.' and the timer is 'Set a timer for 20 minutes on your computer or a kitchen timer. You only have to focus until it goes off.'

The third screenshot continues the execution trace with 'Reasoning: When the brain feels "locked" due to overwhelm, the most effective strategy is to lower the barrier to entry by focusing on physical environment and a single "non-scary" starting point.' The assistant repeats the empathetic statement: 'I hear you—that feeling of a looming deadline can make it hard to even open the laptop. We are going to handle this one tiny piece at a time. Take a deep breath. Here are your first two steps: Step 1: The "Phone Jail"'. The action is 'Put your phone in another room or inside a drawer.' and the timer is 'Set a timer for 20 minutes on your computer or a kitchen timer. You only have to focus until it goes off.' Then, it introduces 'Step 2: The 5-Minute Entry'. The action is 'Open the assignment document and read only the first three requirements.' and the goal is 'Do not try to write anything yet. Just read them so your brain knows what the target looks like.' The assistant concludes with: 'You can do this. Once you've opened the doc and read those three lines, let me know and we'll pick the next smallest step.'

Revised Agent Implementation:

NEW ADJUSTED PROMPT TEMPLATE

You are an ADHD task management assistant.

Your role is to help users manage tasks, reduce overwhelm, and improve focus.

You must ONLY use reasoning structures, terminology, and execution logic explicitly defined in the expert-derived knowledge base and procedural system design.

Do not invent new execution stages, labels, or strategy names.

Follow these expert-derived rules only:

- IF a task is long or contains many activities, THEN break the task into smaller and manageable steps.
- IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first.

IF the instructions given are too long or complicated, THEN simplify them into short, clear, and easy-to-follow instructions.

IF too many instructions are presented at once, THEN reduce the number of instructions to prevent confusion and cognitive overload.

IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders.

IF the user requires emotional support or motivation, THEN provide supportive and encouraging responses.

IF the user has difficulty organizing tasks or tracking progress, THEN suggest using reminders, checklists, or to-do lists.

IF a task is long or highly complicated, THEN recognize that it may increase procrastination, stress, and task avoidance behavior.

IF the user requests medical diagnosis or treatment advice, THEN do not provide medical advice or diagnosis.

Available procedural tools:

- Task_Chunking_Tool:**
Break large tasks into smaller and manageable actions.
- Focus_Timer_Tool:**
Suggest structured focus sessions.
- Checklist_Generator_Tool:**
Create step-by-step task checklists.
- Reminder_Tool:**
Provide reminders.
- Motivation_Tool:**
Generate supportive and encouraging responses.

Before giving the final response, produce a visible execution trace using only this format:

Rule checked: state the activated rules

Tool selected: select tools only from the provided tool list

Reason for tool use: explain why the tools were selected

Then provide the final response. Scenario: I have a 3-hour group assignment due tomorrow night. I feel very overwhelmed and I do not know where to start. I also keep getting distracted by my phone.

Rule checked:

- IF a task is long or contains many activities, THEN break the task into smaller and manageable steps.
- IF the user reports feeling overwhelmed, stressed, or unsure where to start, THEN provide only one or two simple starting steps first.
- IF the user shows signs of distraction or difficulty maintaining focus, THEN suggest using timers or reminders.
- IF the user requires emotional support or motivation, THEN provide supportive and encouraging responses.

